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1-D:

A) The distributors who sell the electricity do not bear the external costs. They only bear “private costs”

B) The electricity company that produces the electricity may bear private costs, such as the cost of **fuel, maintenance, and operation**. However, they do not directly bear the external costs.

C) The government that subsidizes the production of electricity does not directly bear the external costs. While they may provide financial support to the electricity company, they do not experience the negative impacts of electricity generation.

D) The people living in the area where the electricity is generated bear the external costs. They are the ones who suffer the negative impacts of electricity generation, such as air and water pollution, noise, and health effects. They may also face decreased property values, reduced quality of life, and other economic and social impacts. Therefore, it is the people living in the area who bear the external costs of electricity generation. **This explains why key D is correct.**

2-A:

When a profit maximizing firm is allocatively efficient, this represents “marginal cost pricing” being done. This means that **Marginal cost = Average revenue (price)**

This represents that key A will be correct.

B – When marginal cost is equal to Average cost, this represents that Average total cost has been minimized. This represents “productive efficiency”.

C and D: MC does not have to be equal to TR or TC for allocative efficiency to be achieved.

3-C:

A) Government expenditure on enterprise education can actually increase dynamic efficiency. By providing training and support for entrepreneurs and businesses, the government can help foster a culture of innovation, leading to new ideas, products, and processes.

B) Reinvestment of profits can also increase dynamic efficiency. When businesses reinvest their profits, they are able to fund research and development, innovation, and capital expenditures, which can lead to new products, processes, and productivity improvements.

C) Rising shareholder dividends, on the other hand, can reduce dynamic efficiency. When companies prioritize dividend payments over reinvestment, they may have fewer **retained profits available with them** to invest in research and development, innovation, and capital expenditures. **This will subsequently prevent dynamic efficiency from being achieved.**

D) An increase in innovation can directly increase dynamic efficiency. Innovation leads to new products, processes, and productivity improvements, which can drive **dynamic efficiency**.

4-C:

Despite high sales taxes and anti-smoking campaigns, the quantity of cigarettes demanded may remain high because **many smokers are addicted to nicotine and find it difficult to quit**. Addiction can lead to **irrational consumer behavior**, where smokers continue to purchase cigarettes despite the health risks and increased cost. **This explains why key C is correct.**

A) This statement suggests that people are aware of the health risks and are changing their behavior by quitting or reducing smoking. However, this would lead to a decrease in demand, not a high quantity demanded.

B) Since some smokers may switch to cheaper and less dangerous alternatives such as: nicotine patches and gums, this will result in a decrease in quantity demanded of cigarettes.

D) While a price increase may make consumers worse off, it would lead to a decrease in demand, not a high quantity demanded.

5-D:

➔ A budget line, also known as a **budget constraint**, shows the various combinations of two goods that a consumer can purchase given their income and the prices of the goods.

Suppose a consumer has a monthly income of \$100 and wants to buy two goods, X and Y. The prices of X and Y are \$10 and \$5, respectively.

Here's a table showing the various combinations of X and Y that the consumer can buy with their \$100 income:

X (units)	Y (units)	Total Cost
0	20	$\$0 + \$100 = \$100$
2	16	$\$20 + \$80 = \$100$
4	12	$\$40 + \$60 = \$100$
6	8	$\$60 + \$40 = \$100$
8	4	$\$80 + \$20 = \$100$
10	0	$\$100 + \$0 = \$100$

A) The behavior of a utility-maximizing consumer is represented by the consumer equilibrium.

B) The difference between an individual's consumption and saving is not represented by the budget line.

C) The monetary value of the disposable income of a consumer is also not represented by the budget line.

6-C:

Key take-away: Giffen goods prove law of demand incorrect.

A Giffen good is a unique case where the income effect and substitution effect work in opposite directions, and the income effect is stronger, **or in other words, the income effect outweighs the substitution effect.**

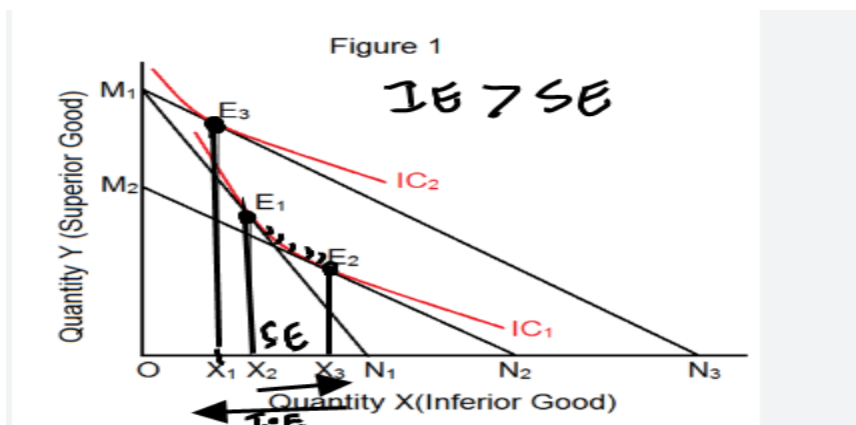
- The substitution effect: When the price falls, consumers tend to buy more of the good (normal behavior).
- The income effect: However, the price fall results in consumer's real income to rise sufficiently such that they contract their qty demanded of the giffen good (that is loaf) and switch to other desirable alternatives that include: fresh fruit and vegetables to make up their diet.

Note:-Some common examples of Giffen goods include:

1. Staple foods like bread, rice, or potatoes in poor households.
2. Low-quality housing or slum dwellings.
3. Cheap, low-quality clothing or fabrics.

These goods are often considered inferior because people tend to switch to better alternatives as their real income increases. For instance, someone who can afford to buy more nutritious food will likely switch from staple foods like bread and potatoes to healthier options like fruits, vegetables, and lean proteins.

In a Giffen good, the income effect is stronger, causing the quantity demanded to decrease when the price falls. **This represents that they prove “law of demand incorrect”**



This explains why key C is correct.

No, the income effect is not irrelevant for a giffen good; instead, it is the most important. (A)

When the income effect works in the same direction as the substitution effect, this represents the case of normal goods. (B)

When the income effect works against the substitution effect but is of a smaller magnitude, this represents inferior goods. (D)

7-C:

- The high **five-firm concentration ratio** (65%) indicates that a small number of firms (5) control a significant share of the market, which is a characteristic of **oligopoly**.
- The presence of many firms (250) and brands (1000) suggests that there is some competition, but the concentration ratio suggests that it is not perfect competition.
- Monopoly is ruled out since there are multiple firms.
- Monopolistic competition is also unlikely since the concentration ratio is high, indicating a more concentrated market.

Option	Characteristics	Reasoning
Monopoly	Single firm, entire market	250 firms, 1000 brands → Not a monopoly
Monopolistic Competition	Many firms, differentiated products	High concentration ratio (65%) → Not monopolistic competition
Oligopoly	Small number of firms, interdependent decisions	High concentration ratio (65%), 5 firms dominate → Oligopoly likely
Perfect Competition	Many firms, homogeneous product, no barriers	High concentration ratio (65%), only 250 firms → Not perfect competition

8-C: **KEY TAKEAWAY: Sales (volume) maximization is ($ATC=AR$) whereas sales revenue maximization ($MR=0$)**

By setting the price at the highest point that ensures all tickets will be sold, they are prioritizing selling out the concert over maximizing profits. **This represents that the objective of “sales maximization” (C) is kept.**

A) Profit maximisation: Their primary goal is to sell all tickets, not maximize profits.

B) Revenue maximisation: The band's focus is on selling all tickets, not maximizing revenue.

D) Satisficing: This term refers to a decision-making strategy that aims for "good enough" outcomes, rather than optimal ones. In this case, the band is aiming to sell all tickets, which suggests they are striving for optimality, not just satisficing.

9-B:

B) Personal services are typically more labor-intensive and less capital-intensive, making it easier for small firms to compete in these industries. An increase in personal services as a share of GDP would likely lead to more opportunities for small firms to survive and thrive.

A) An increase in the cost of borrowing from commercial banks would make it harder for small firms to access capital, making survival more difficult.

C) An increase in the rate of change of information technology could lead to economies of scale that favor larger firms, making it harder for small firms to compete.

D) An increase in the ratio of capital to labor employed in an economy would also favor larger firms that have more access to capital, making it harder for small firms to survive.

10-D:

A cartel is a group of firms that collude to restrict output and raise prices. For a cartel to be successful, it needs to be able to enforce its agreements among member firms. Strictly enforced production quotas are essential to ensure that each firm produces only its agreed-upon share, preventing any individual firm from cheating and undermining the cartel's price-fixing efforts. **This explains why key D is correct.**

A) A large number of firms in the industry would actually make it harder to maintain a cartel, as it would be more difficult to coordinate and enforce agreements among many firms.

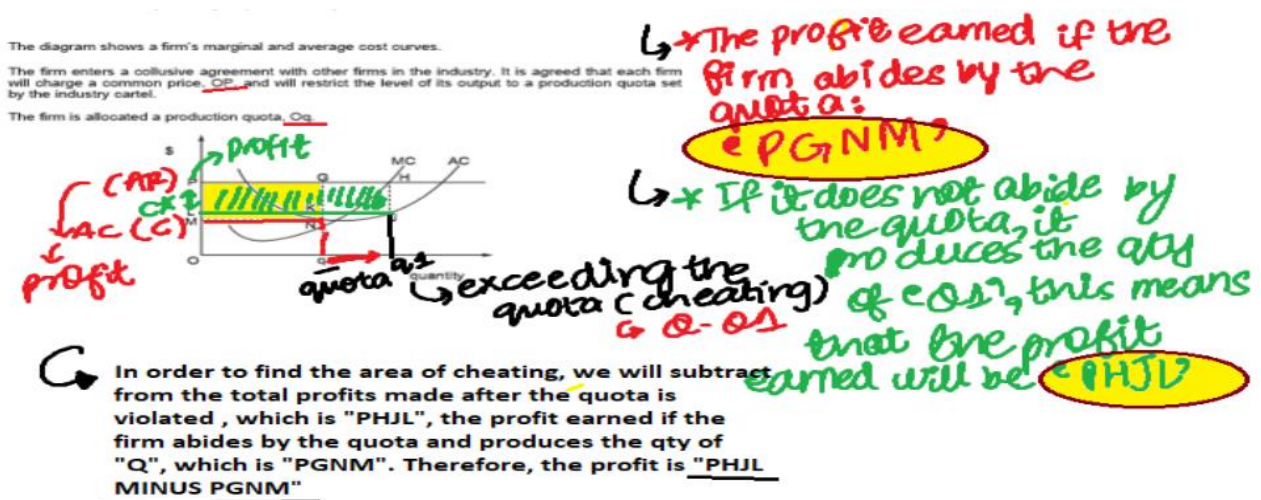
B) Having a differentiated product would not contribute to a successful cartel, as differentiation can lead to competition among firms. (usually seen in non-collusive oligopoly)

C) Low barriers of entry to the industry would actually threaten a cartel's survival, as new firms could enter the market and disrupt the cartel's control over prices and output.

11-D:

- Diseconomies of scale in the combined company before the separation: Yes, the company may have been experiencing diseconomies of scale, making it harder to manage and operate the combined business.
- Markets can easily be separated by this company: Yes, the company can separate its international and domestic operations into distinct businesses.
- The level of competition and price elasticity in domestic markets: Yes, the company faces less competition in domestic markets, which have a lower price elasticity of demand, making it a more attractive market to operate in.

12- C:



13-D:

The correct answer is D) Too many households own cars so the market in permits is very small.

The policy relies on the creation of a market for parking permits, where households with extra permits can sell them to households with multiple cars. However, if too many households own cars, the demand for permits will be very high, and the supply of permits from households with no cars will be very low, making the market for permits very small. This could lead to a situation where the price of permits is very high, making it difficult for households with multiple cars to purchase them, and ultimately leading to the failure of the policy.

A) Fines for drivers parking in the area without a permit would actually complement the permit system and help internalize the externality.

B) The local government's monopoly over the issue of parking permits is not a reason for the policy to fail. In fact, it allows the government to control the supply of permits and create a market for them.

C) The policy is designed to compensate households without cars by allowing them to sell their permits to households with multiple cars, so this is not a reason for the policy to fail.

14-B:

Nudge theory aims to influence people's behavior in predictable ways without limiting their freedom of choice. Nudges are subtle changes in the environment that guide people toward making better choices.

B) Imposing a tax on sugary snacks is a policy tool that uses economic dis-incentives to change behavior, rather than a nudge.

The other options are all examples of nudges:

A) The coffee shop is using a nudge by offering a reward (a larger coffee) for bringing a reusable cup, encouraging sustainable behavior.

C) The opt-out system for organ donation is a nudge, as it changes the default option to increase donations.

D) The restaurant chain is using a nudge by introducing smaller main courses, making it easier for customers to choose a smaller portion and reduce food waste.

15-B:

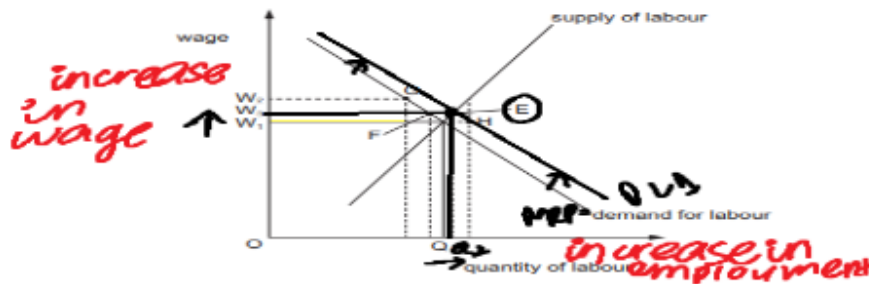
A - Direct taxes, such as income taxes, are typically progressive, meaning:

- The Average tax rate increases as income rises
- The marginal tax rate (MRT) is higher than the average tax rate (ART)

This means that as income increases, the tax rate applied to each dollar earned (MRT) is higher than the overall average tax rate (ART). **This progressive structure aims to reduce income inequality by taxing higher incomes at a higher rate.**

C- If wage is kept at W_1 , and the equilibrium is of H, then although there will be an increase in employment, there will not be an increase in wages.

D – If there is an increase in marginal revenue productivity, or in other words, demand for labor and a new equilibrium of “E” is achieved, there will be a rise in wages and employment, as can be seen below. This is why, this becomes correct.



18-C:

Sustainable economic growth refers to a growth path that balances the needs of the present with the needs of the future. It means achieving economic growth while ensuring that the natural resources and environmental assets are preserved for future generations. **In other words, sustainable economic growth meets the needs of the present without compromising the ability of future generations to meet their own needs.**

Option C captures this idea by emphasizing the importance of maintaining future productive potential while pursuing current economic growth.

A) Increasing productive potential without ensuring that the future productive potential does not reduce would not lead to **sustainable growth**.

B) Using only low-cost resources may lead to overreliance on finite resources, ignoring the need for long-term sustainability.

D) Focusing solely on increasing future productive potential may neglect the needs of the present generation.

19-D:



➔ $MEW = GNP + \text{non-marketed activities} + \text{value of leisure} - \text{pollution (environmental damage)} - \text{crime rate}.$

Now, let's say Country B has a GNP of \$90 billion, but:

- Leisure time is valued at \$12 billion
- Unmarketed activities are valued at \$20 billion
- Pollution costs are \$3 billion
- Crime rate costs are \$1 billion

MEW would be:

$$\begin{aligned} \text{MEW} &= \text{GNP} + \text{Leisure} + \text{Unmarketed activities} - \text{Pollution} - \text{Crime rate} \\ &= \$90 \text{ billion} + \$12 \text{ billion} + \$20 \text{ billion} - \$3 \text{ billion} - \$1 \text{ billion} \\ &= \$118 \text{ billion} \end{aligned}$$

The MEW is greater than GNP because Country B has a higher value of non-marketed goods and services and leisure, which increases its MEW.

This explains why key D will be correct.

A) If there were negative externalities such as: pollution, this will reduce the value of MEW. (MEW= GNP- value of negative externalities)

B and D: Property income from abroad and regrettable necessities are included in the GNP.

20-B:

country	life expectancy at birth (years)	Gross National Income per capita (\$)	expected years of schooling	skilled labour (% of labour force)
Croatia	78.3	23 061	15.0	91.5
Norway	82.3	68 059	18.1	84.3
United Arab Emirates	77.8	66 912	13.6	52.8
Qatar	80.1	110 489	12.2	43.9

- Life expectancy at birth: Norway has the **highest life expectancy (82.3 years)**, indicating a high level of healthcare and living standards.
- Gross National Income per capita: Norway has the **second highest GNI per capita (\$68,059)**, indicating a strong economy and high standard of living.
- Expected years of schooling: Norway has **the highest number of expected years of schooling (18.1)**, indicating a strong education system and investment in human capital.
- Skilled labor force: Norway has the **second highest percentage of skilled labor force (84.3%)**, indicating a highly educated and trained workforce.

This explains why key B is correct.

21-D:

A) Higher taxes can reduce disposable income, leading to lower consumer spending. This can dampen Aggregate demand, negatively impacting economic growth.

B) A high birth rate represents that there will be a greater number of dependents which will increase the dependency ratio. In fact, a rapidly growing population can put pressure on resources, infrastructure, and services, potentially hindering economic development.

C) Achieving a budget surplus often requires austerity measures, like reducing government spending or increasing taxes. These measures can reduce aggregate demand, potentially harming economic growth.

D) Training and up-skilling the workforce can improve productivity. This can lead to increased economic output and growth. By upgrading the skills of their workforce, developing countries can unlock their economic potential and drive growth. **It will boost the Long run Aggregate Supply and productive potential in the long-term i.e. over the next five years.**

22-D:

D) The quote suggests that if people are incentivized not to work (through payments or benefits) and are taxed when they do work, they may choose not to work, leading to voluntary unemployment. **This is an example of voluntary unemployment, as individuals are choosing not to work due to the incentives and tax structure.**

This explains why key D is correct.

A) Cyclical unemployment refers to unemployment caused by economic downturns or recessions.

B) Frictional unemployment occurs when workers are temporarily unemployed while searching for new jobs or transitioning between jobs.

C) Structural unemployment results from changes in the pattern of demand and supply in the economy.

23- B:

Unemployment leads too:-

- A) A fall in aggregate demand, as fewer people have less income to spend.
- C) A fall in revenues from income tax, as fewer people are working and paying taxes.
- D) A fall in the rate of economic growth as unemployment results in reduced spending which slows down Aggregate demand and economic activity.

However, a rise in unemployment would likely lead to an increase in expenditure on welfare, such as unemployment benefits and other forms of support, as more people become eligible and require assistance. **Therefore, option B is the least likely consequence.**

During economic recessions, governments often face increased pressure on welfare systems and increased spending in this area to support those affected by unemployment.

24-C:

Total Currency Flow refers to the total amount of money flowing between domestic residents (individuals, businesses, and banks) and foreign residents (foreign banks, governments, and other entities). **This explains why key C will be correct.**

25-D:

This transaction does not change the level of withdrawals from the circular flow of income because it is merely a transfer of funds within the financial sector, not a withdrawal from the circular flow. The money is still being held within the banking system and can be used for future transactions. **This explains why key D will be correct.**

A) An increase in income tax rates will result in higher **withdrawals or leakages from the circular flow.**

B) Spending more on imported cars following a currency appreciation represents an increase in imports, which is a **withdrawal from the circular flow.**

C) Withdrawing savings to spend on a celebration **represents a reduction in withdrawals since savings is a withdrawal.**

26-C:

The correct answer is **C) Unemployment can still persist in the long run.**

Keynesians argue that:

- Unemployment (**specifically demand-deficient unemployment**) can persist even in the long run, due to lack of Aggregate demand.
- Government intervention is necessary to stabilize output and employment.
- Fiscal policy (government spending and taxation) is a key tool for achieving this goal.

Instead, monetarists argue that in the long-run, an economy operates on its potential and there is only natural rate of unemployment.

A) Keynesians do not prioritize balanced budgets; instead, they advocate for deficit spending to stimulate the economy during downturns. **Rather, monetarists advocate for balanced budgets.**

B) According to monetarist, money supply has a direct impact on the rate of inflation. **Remember the quantity theory of money.**

D) Keynesians argue that wages are sticky, not flexible, in the short run, which can lead to unemployment persisting even in the face of downward pressure on wages. **Instead, monetarists believe that wages are flexible downwards.**

Topic	Keynesian Approach	Non-Keynesian Approach
Balanced Budgets	Not a priority Deficits can be necessary for economic stimulus	Priority Balanced budgets are essential for economic stability
Inflation	Demand-side factors (aggregate demand, consumption) Fiscal policy (govt spending, taxation) to address inflation	Excessive money supply growth Monetary policy (central bank control) to combat inflation
Unemployment	Can persist in the long run Government intervention necessary to address unemployment	Short-term phenomenon Economy will naturally return to full employment
Wages	Sticky (don't adjust quickly to market conditions) Unemployment can persist even with high wages	Flexible (adjust quickly to market conditions) Full employment will be achieved through wage adjustments

27-D:

In-order to offset the impact of an increase in leakage:

- ➔ Either leakages need to be decreased.
- ➔ Or, injections need to be increased.

This is because a macro-economic equilibrium is established where: J (injections) = W (leakages)

- A) A reduction in domestic consumption because of a rise in savings will further widen the disequilibrium.
- B) A reduction in government expenditure which represents an injection into the circular flow will widen the disequilibrium.
- C) An increase in the rate of interest after having increased the costs of borrowing will reduce investment, which is an injection into the circular flow, thereby further widening the disequilibrium.
- D) An increase in investment which represents an injection into the circular flow will help to offset an impact of an increase in leakage, and the economy will restore its equilibrium whereby "W=J"

28-D:

When life expectancy falls, but the birth rate remains high, it means that:-

- More children are being born
- People are living shorter lives

As a result, the working-age population (those who are productive and contributing to the economy) will shrink as a percentage of the total population. This is because there will be more dependents (children and younger population) and fewer working-age individuals to support them. **As a result of which, key D, becomes correct.**

A) Fewer schools and resources for young people will not be needed, as the birth rate is high, indicating a larger younger population.

B) More facilities for older people may not be needed, as the life expectancy has fallen, indicating a shorter average lifespan.

C) The population will not be aging in this scenario, as the birth rate is high, and the population will be relatively young.

29-B:

B) The Laffer curve suggests that as tax rates increase, tax revenue initially increases, but eventually decreases as high tax rates discourage economic activity and lead to tax evasion or avoidance. **The economist's prediction that increasing the top tax rate from 40% to 45% will reduce tax revenue by 2% is consistent with the Laffer curve's idea that higher tax rates can lead to lower tax revenue.**

A) Kuznets curve: relates to income inequality and economic growth, not tax rates and revenue.

C) Lorenz curve: visualizes income distribution, not tax revenue.

D) Phillips curve: represents the trade-off between unemployment and inflation, not tax rates and revenue.

30-B:

This combination of policies is most likely to reduce the **rate of inflation** because:

All these are **contractionary or deflationary policies**.

- Decreasing government spending reduces aggregate demand, which helps reduce inflation.
- Increasing interest rates makes borrowing more expensive, which reduces investment, because of which, **Aggregate demand reduces**. This eventually causes inflation to reduce.
- Increasing direct taxation on households reduces disposable income, leading to reduced consumption and demand for goods and services, thus helping to reduce inflation.

